AP Calculus AB

Course Outline • Mathlify Summer Program • College Calculus I Preparation

Course Overview

AP Calculus AB is equivalent to College Calculus I. The course develops understanding of limits, derivatives, and integrals through graphical, numerical, analytical, and verbal representations. Students are prepared to enter the school year confident, organized, and ready to succeed on the AP exam.

Course Units

Unit 1 — Limits & Continuity

• Limits from graphs, tables, and algebra

• One-sided limits and limits at infinity

• Continuity and Intermediate Value Theorem

• Asymptotic and end behavior of functions

Unit 2 — Derivatives: Definition & Rules

• Derivative as a limit of the difference quotient

• Power, product, quotient, and chain rules

• Trig, inverse trig, exponential, and logarithmic derivatives

• Implicit differentiation

Unit 3 — Applications of Derivatives

• Tangent lines and rates of change

• Critical points, extrema, and concavity

• First and second derivative tests

• Optimization and related rates problems

• Mean Value Theorem and Rolle's Theorem

Unit 4 — Integrals & Accumulation

• Riemann sums and definite integrals

• Area under and between curves

• Fundamental Theorem of Calculus

• Trapezoidal Rule and accumulation functions

Unit 5 — Integration Techniques

• Antiderivatives and initial value problems

• Substitution (u-substitution)

• Inverse trig and logarithmic integrals

Unit 6 — Differential Equations & Slope Fields

• Slope fields and solution curves

• Separable differential equations

• Exponential growth and decay models

• Newton's Law of Cooling

Unit 7 — Applications of Integrals

• Volumes by slicing

• Disk and washer methods

• Volumes with known cross sections

Unit 8 — L'Hôpital's Rule & Improper Integrals

• Improper integrals and convergence

• Understanding L'Hôpital's Rule and indeterminate forms

• Structured free-response justification

Assessment & Practice

Students complete regular AP-style multiple choice and free-response questions. Emphasis is placed on clear written justification, multiple representations, and strategic calculator usage.